Replication Package for “Two-sided Search in International Markets”

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This package contains the code, data files, saved parameter vectors, and generated outputs needed to replicate the Journal of Political Economy manuscript in docs/EJTX_2026_replication.pdf, with source in docs/EJTX_2026_replication.tex.

1. Data Availability

The main microdata source is the U.S. Census Bureau Longitudinal Firm Trade Transactions Database (LFTTD), accessed through Federal Statistical Research Data Center project 1109. These confidential shipment-level data are restricted by law and are not included in this public package.

The package does include the statistics we generated from the LFTTD that are needed for public replication of our structural model estimates and counterfactuals. These statistics were cleared by the Census Bureau’s Disclosure Review Board and Disclosure Avoidance Officers under requests 5488-2017-0109, 5623-2017-0228, 5952-2017-0808, and 11212-2024-0229.

The package also includes the Stata code that generated the LFTTD-based disclosed statistics (in `` `code/Stata_LFTTD_code/` ``), as well as public data from the Bureau of Economic Analysis, World Trade Organization, Department of Commerce, Census Annual Retail Trade Survey, and U.S. International Trade Commission.

2. Statement About Rights

I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package. Appropriate permissions are documented in `LICENSE-CODE.txt` and `LICENSE-DATA-DOCS.txt`, as applicable.

3. Package Contents

- `README.pdf`: main replication instructions.
- `LICENSE-CODE.txt`: MIT license for code files.
- `LICENSE-DATA-DOCS.txt`: CC-BY 4.0 license for included data and documentation, except where original sources impose different terms.
- `docs/EJTX_2026_replication.pdf: replication manuscript used for the public package.`
- `docs/EJTX_2026_replication.tex: LaTeX source for the replication manuscript.`
- `docs/table_figure_source_map.xlsx`: map from each paper and appendix table/figure to its code and data source.

- `code/matlab/`: MATLAB code for model solution, table generation, counterfactuals, figures, exact-fit checks, and standard errors.
- `code/Stata_LFTTD_code/`: Stata code used inside the FSRDC to construct restricted Census-derived statistics.
- `code/Stata_figure11b_event_study/`: Stata code for Figure 11b.
- `data/BEA_and_WTO--trade_and_production_aggregates/`: public data for Figure 1.
- `data/Census_LFTTD--disclosed_statistics_and_graphs/`: disclosure-approved Census-derived files for Figures 2, 3, and 6 and Tables 1-4.
- `data/Dept_of_Commerce--apparel_imports_by_origin/`: public data for Figure 4.
- `data/Census_ARTS--apparel_retailer_aggregates/`: public ARTS data used for apparel retailer cost/revenue calibration.
- `data/US_ITC_figure11b_event_study/`: event-study inputs for Figure 11b.
- `data/intermediate/`: compact full-precision MP appendix parameter file.
- `output/`: generated paper-facing tables, figures, and logs.

4. Software

The package was prepared on Linux. The public replication uses:

- MATLAB R2025b with Global Optimization Toolbox and Parallel Computing Toolbox.
- Stata for Figure 11b, with SSC packages `ebalance`, `ftools`, `require`, and `reghdfe`.
- LibreOffice only if regenerating Word/Excel documentation files.

Optional tools for documentation rebuilds are LibreOffice and pandoc.

5. Public Replication Run

Start MATLAB from the package root and run:

```
cd('/path/to/replication_package')
addpath(genpath(fullfile(pwd, 'code', 'matlab')))
run_all_replication('fast')
```

The fast run regenerates or collects the paper-facing outputs in:

- `output/figures/paper/`
- `output/tables/paper/`
- `output/logs/`

The fast run does not reestimate the structural models. Instead, it loads the paper-reported or saved best-fit parameter vectors, evaluates the model code at those vectors, regenerates standard errors for Tables 5, 9, and 10, and recreates the paper-facing tables and figures.

Figure 11b is generated by Stata. To run it directly from the package root:

```
ssc install ebalance, replace
ssc install ftools, replace
```

```
ssc install require, replace
ssc install reghdfe, replace
do code/Stata_figure11b_event_study/event_study.do
```

6. Expected Running Time

Expected running time on the Linux workstation used to prepare the package:

- `run_all_replication('fast')`: about 8-10 minutes.
- `setenv('EJTX_TEST_RUN','1'); run_all_replication('full')`: about 15-20 minutes. This checks full-estimation entry points without running optimizers to completion.
- `run_all_replication('full')` without `EJTX_TEST_RUN`: several days to multiple weeks, depending on hardware and toolbox performance.

The `full` mode is not required to reproduce the paper tables and figures because the package includes the reported best-fit parameter vectors and exact-fit checks.

7. Restricted Census Code

The raw LFTTD data cannot be distributed. Approved researchers can run the Census construction scripts inside an FSRDC.

Inside the FSRDC, copy `code/Stata_LFTTD_code/` to the project program directory and edit the path globals at the top of `00_do_apparel_shell_abbrev.do`:

```
global Data       "/path/to/restricted/source/data"
global Dofile     "/path/to/code/Stata_LFTTD_code"
global Output     "/path/to/private/generated/data"
global OutputTemp "/path/to/private/generated/temp"
global Result     "/path/to/private/disclosure/request/files"
global OutputLog  "/path/to/private/logs"
```

Then run:

```
do "$Dofile/00_do_apparel_shell_abbrev.do"
```

The Census Bureau has updated raw LFTTD file names since the code was originally run, so users inside an FSRDC may need to update raw-data file references. Outputs from the restricted environment must pass Census disclosure review before they can be moved into the public package.

8. Output Checks

The recommended fast run writes:

- `output/logs/run_all_replication_fast.log`
- `output/logs/exact_fit_checks.log`
- `output/logs/standard_error_generation.log`

- `output/tables/paper/model_fit_checks_for_tables_05_09_10_11_12.csv`
- `output/tables/paper/generated_standard_errors_at_saved_parameters.csv`

The exact-fit checks evaluate the packaged MATLAB model code at the reported/saved parameter vectors and report saved-versus-regenerated objective values. The standard-error log reports regenerated standard errors for the relevant model tables.

9. Data Citations

- U.S. Census Bureau, Longitudinal Firm Trade Transactions Database and related Census business microdata, accessed through FSRDC project 1109.
- U.S. Census Bureau, disclosure-approved output files for approval requests 5488-2017-0109, 5623-2017-0228, 5952-2017-0808, and 11212-2024-0229.
- U.S. Census Bureau, Annual Retail Trade Survey.
- Bureau of Economic Analysis, domestic apparel production series.
- World Trade Organization, apparel trade series.
- U.S. Department of Commerce, apparel imports by origin.
- U.S. International Trade Commission, Section 301 tariff coverage and apparel-sector impacts.

10. Omitted Materials

The only omitted materials are confidential Census microdata and non-disclosable derivatives. The public package includes the disclosure-approved statistics and artifacts needed to reproduce the public-facing model estimates, fit checks, figures, tables, and counterfactuals.